Name _____ Period _____ Date _____

3-3 Lesson Quiz: Polynomial Identities

short form questions keep their Savvas numbers (2, 4, 5); show your work

2. Part A Use polynomial identities to multiply $(5 - 4x^3)(5 + 4x^3)$.

Ⓐ $25 - 4x^9$

Ⓑ $25 - 40x^3 + 16x^6$

Ⓒ $25 - 4x^6$

Ⓓ $25 - 16x^6$

Part B What identity did you use?

Ⓐ Difference of squares

Ⓑ Difference of cubes

Ⓒ Sum of cubes

Ⓓ Square of a sum

Work:

4. Use Pascal's Triangle to expand the expression $(x + 2)^8$.

Ⓐ $x^8 + 256$

Ⓑ $256 + x + 16x^2 + 112x^3 + 448x^4 + 1120x^5 + 1792x^6 + 1792x^7 + 1024x^8$

Ⓒ $x^8 + 16x^7 + 112x^6 + 448x^5 + 1120x^4 + 1792x^3 + 1792x^2 + 1024x + 256$

Ⓓ $x^7 + 14x^6 + 84x^5 + 280x^4 + 560x^3 + 672x^2 + 448x + 128$

Show your work. Row 8 of Pascal's Triangle: _____

Work:

5. Expand $(a + 4)^5$:

$$a^5 + _____\, a^4 + _____\, a^3 + _____\, a^2 + _____\, a + _____$$

Work:

Scoring: Question 4 and Question 5 are 10 points each. Question 2 is a bonus of up to 5 points. Total out of 20.

3-3 Lesson Quiz, short form: answer key

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2. **Part A D, $25-16x^6$; Part B A, Difference of squares.** $(5-4x^3)(5+4x^3) = 5^2-(4x^3)^2 = 25-16x^6$. Choice C squares the variable but not the coefficient 4 (the TE's Try It 2 common error); A also multiplies the exponents $3 \cdot 3$; B is $(5-4x^3)^2$, a square, not a product of a sum and a difference. Plain distribution (Lesson 3-2) gets the same product. Savvas quiz item 2, DOK 2, Skills Review and Practice G100. Taught by TE Example 2 (cut in the fast packet; the packet's reference box names the identity). **Bonus 5:** 3 for Part A, 2 for Part B.

4. **C.** Row 8 is $1, 8, 28, 56, 70, 56, 28, 8, 1$; with $a = x$, $b = 2$ the terms are x^8, $8(2)x^7 = 16x^7$, $28(4)x^6 = 112x^6$, $56(8)x^5 = 448x^5$, $70(16)x^4 = 1120x^4$, $56(32)x^3 = 1792x^3$, $28(64)x^2 = 1792x^2$, $8(128)x = 1024x$, $2^8 = 256$. Choice A keeps only the first and last terms; B attaches the coefficients to the wrong powers of x; D is $(x+2)^7$ (Row 7). Savvas quiz item 4, DOK 2, G100. Taught by Example 5A and Try It 5a. **10 points:** 6 for C, 4 for a correct Row 8 or shown substitution.

5. **20; 160; 640; 1280; 1024.** Row 5 is $1, 5, 10, 10, 5, 1$; $5(4) = 20$, $10(4^2) = 160$, $10(4^3) = 640$, $5(4^4) = 1280$, $4^5 = 1024$. Savvas quiz item 5, DOK 2, G100. Taught by Example 5B and Try It 5b. **10 points:** 2 per blank.

Items left off this form (available for a retake or Form B): Savvas item 1, prove $x^3 - y^6 = (x-y^2)(x^2+xy^2+y^4)$ by filling in the four step reasons from a word bank (answer: Distributive; Multiply; -1; like terms; DOK 2, G100; TE Example 1, cut); Savvas item 3, factor $1-125n^3$, select all factors, and name the identity (answer: Part A A and E, $1-125n^3 = (1-5n)(1+5n+25n^2)$; Part B B, Difference of cubes; DOK 2, G100; TE Example 3, cut). Item 3 is the same skill as Topic 3 assessment item 7 (factor $64+27a^3$) and IXL NJV (an optional, unstarred row in the packet's IXL table); if the PLC adds Example 3, move item 3 onto this form. The quiz has no graphs, so nothing was left off for a figure.

IXL replacement: SmartScore 70 or above on A7M and NEU replaces this quiz with 20/20.